

SAT Math

Nonlinear Equations and Systems 1

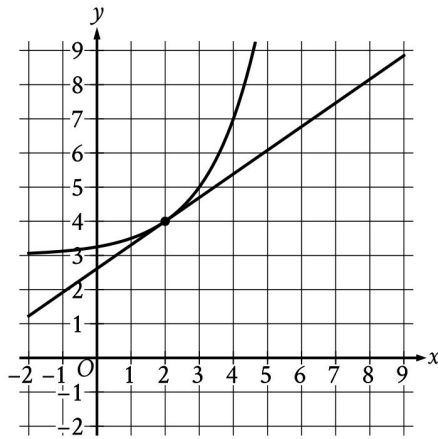
Question # ID
1.1 3c95093c

$$6x - 9y > 12$$

Which of the following inequalities is equivalent to the inequality above?

- A. $x - y > 2$
- B. $2x - 3y > 4$
- C. $3x - 2y > 4$
- D. $3y - 2x > 2$

1.2 4ca30186



The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution (x, y) to this system?

- A. $(0, 0)$
- B. $(0, 2)$
- C. $(2, 4)$
- D. $(4, 0)$

1.3 3de7a7d7

Which of the following is a solution to the equation $2x^2 - 4 = x^2$?

- A. 1
- B. 2
- C. 3
- D. 4

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Question # ID

1.4 70f98ab4

$$q - 29r = s$$

The given equation relates the positive numbers q , r , and s . Which equation correctly expresses q in terms of r and s ?

A. $q = s - 29r$

B. $q = s + 29r$

C. $q = 29rs$

D. $q = -\frac{s}{29r}$

1.5 568aaf27

$$x + y = 12$$

$$y = x^2$$

If (x, y) is a solution to the system of equations above, which of the following is a possible value of x ?

A. 0

B. 1

C. 2

D. 3

1.6 b76a2815

$$P = \frac{W}{t}$$

The power P produced by a machine is represented by the equation above, where W is the work performed during an amount of time t . Which of the following correctly expresses W in terms of P and t ?

A. $W = Pt$

B. $W = \frac{P}{t}$

C. $W = \frac{t}{P}$

D. $W = P + t$

1.7 c7789423

$$|x - 2| = 9$$

What is one possible solution to the given equation?

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Question # ID

1.8 eb268057

$$x^2 = 64$$

Which of the following values of x satisfies the given equation?

- A. -8
- B. 4
- C. 32
- D. 128

1.9 98f735f2

The total revenue from sales of a product can be calculated using the formula $T = PQ$, where T is the total revenue, P is the price of the product, and Q is the quantity of the product sold. Which of the following equations gives the quantity of product sold in terms of P and T ?

- A. $Q = \frac{P}{T}$
- B. $Q = \frac{T}{P}$
- C. $Q = PT$
- D. $Q = T - P$

1.10 fcb78856

$$b = 42cf$$

The given equation relates the positive numbers b , c , and f . Which equation correctly expresses c in terms of b and f ?

- A. $c = \frac{b}{42f}$
- B. $c = \frac{b-42}{f}$
- C. $c = 42bf$
- D. $c = 42 - b - f$

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Question # ID

1.11 4236c5a3

If $(x + 5)^2 = 4$, which of the following is a possible value of x ?

- A. 1
- B. -1
- C. -2
- D. -3

1.12 f11ffa93

$$\sqrt{x + 4} = 11$$

What value of x satisfies the equation above?

1.13 5639dd1a

$x^2 = (22)(22)$ What is the positive solution to the given equation?

1.14 c1964c11

$$p + 34 = q + r$$

The given equation relates the variables p , q , and r . Which equation correctly expresses p in terms of q and r ?

- A. $p = q + r + 34$
- B. $p = q + r - 34$
- C. $p = -q - r + 34$
- D. $p = -q - r - 34$

1.15 7cb3a8ee

$$|x - 5| = 10$$

What is one possible solution to the given equation?

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Question # ID

1.16 b8c4a1cd

$$8j = k + 15m$$

The given equation relates the distinct positive numbers j , k , and m . Which equation correctly expresses j in terms of k and m ?

A. $j = \frac{k}{8} + 15m$

B. $j = k + \frac{15m}{8}$

C. $j = 8(k + 15m)$

D. $j = \frac{k+15m}{8}$

1.17 7a8cb72a

$$7m = 2(n + p)$$

The given equation relates the positive numbers m , n , and p . Which equation correctly gives m in terms of n and p ?

A. $m = \frac{2(n+p)}{7}$

B. $m = 2(n + p)$

C. $m = 2(n + p) - 7$

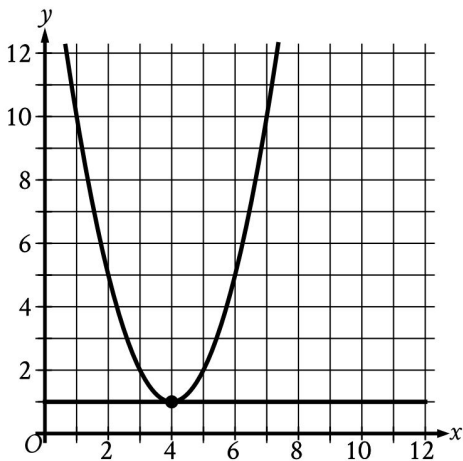
D. $m = 2 - n - p - 7$

Question # ID
1.18 d0e8e8f5

Question ID d0e8e8f5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: d0e8e8f5



- The graph of a system of a linear and a quadratic equation is shown. What is the solution (x, y) to this system?
- A. $(0, 0)$
 - B. $(-4, 1)$
 - C. $(4, -1)$
 - D. $(4, 1)$

1.19 13e5a5d5

$5|x| = 45$

What is the positive solution to the given equation?

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Question # ID

1.20 58443765

$$y = 5x + 4$$
$$y = 5x^2 + 4$$

Which ordered pair (x, y) is a solution to the given system of equations?

- A. $(0, 0)$
- B. $(0, 4)$
- C. $(8, 44)$
- D. $(8, 84)$

1.21 332cd67b

$$3x^2 - 15x + 18 = 0$$

How many distinct real solutions are there to the given equation?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero